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THE STORAGE VIEW OF KV CACHE: CHARACTERIZING CACHE BEHAVIOR ACROSS LLM WORKLOADS

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ABSTRACT

Modern LLM serving systems increasingly extend KV caches beyond GPU memory into host DRAM, SSDs, and remote storage, yet these stores are largely managed using workload-agnostic policies such as LRU and fixed TTLs. To guide the design of KV storage for LLMs, we ask what *execution context* storage needs to make retention, reclamation, and placement decisions across heterogeneous LLM workloads.

We characterize KV-cache reuse, lifetime, and footprint at the granularity of KV-cache blocks across **11 production and public traces spanning six categories**: chat, API serving, RAG, coding agents, tool-using agents, and multi-lineage workflows. We analyze where reuse comes from, when and how frequently KV blocks recur, how long they remain useful, and how they use storage capacity. Across workloads, KV state exhibits two recurring forms. *Single-session state* is reused within one execution: its footprint grows as sessions deepen and accumulates across concurrent sessions, and it can remain useful through long idle periods but loses value when the session completes. *Multi-session state* recurs across sessions of the same workload, outlives individual sessions, and ranges from hot prefixes to a large infrequent tail.

While prior systems often exploit application-, agent-, or workflow-specific information, our results suggest that much of the structure needed for KV-storage management can be exposed through three generic signals: a workload identifier, a session identifier, and a session completion signal. Trace-driven comparisons quantify their value: workload-specific recurrence estimates retain up to 22% more within-session hits and reduce dead state by up to 40%; the session completion signal removes the fundamental tradeoff of idle-time reclamation for single-session state; and session-aware reuse classification identifies shared state 4–700× more selectively than request-level access counts. These results motivate a small cross-layer interface for workload-aware KV storage without requiring application-specific semantics.

1 INTRODUCTION

Large language models (LLMs) now power applications ranging from interactive chat and API services to coding assistants and autonomous agents (Wang et al., 2024; Patel et al., 2024; Yao et al., 2023; Yang et al., 2024). Serving these models efficiently depends heavily on the *KV cache*: the key and value tensors computed for previously processed tokens (Kwon et al., 2023). When requests share a prefix (such as a system prompt, retrieved document, or conversation history), the corresponding KV state can be reused, avoiding redundant prefill computation (Kwon et al., 2023; Zheng et al., 2023; Gim et al., 2024).

Long contexts, large KV footprints, and many concurrent sessions can quickly exhaust GPU memory (Kwon et al., 2023; Lin et al., 2024a; Sheng et al., 2023). Serving systems therefore increasingly extend KV caches into host DRAM, local SSDs, and remote or distributed storage, retaining KV state so that it can be reloaded rather than recomputed (Gao et al., 2024; Yu et al., 2025; Qin et al., 2025; Liu et al., 2025). Systems such as Mooncake (Qin et al., 2025), LMCache (Liu et al., 2025), SGLang HiCache (Zheng et al., 2023), and vLLM KV offloading (Kwon et al., 2023) make KV storage a first-class component of the serving stack.

We use *KV storage* to refer to this multi-tier storage and management layer: the component that retains KV blocks beyond the invocation that creates them, places them across available tiers, and makes them available for reuse. Its central decisions are therefore what state to retain, how long to retain it, and where to place it. Existing KV stores largely make these decisions using generic cache signals such as recency and fixed time-to-live (TTL) (Liu et al., 2025; Qin et al., 2025; Zheng et al., 2023; Kwon et al., 2023).

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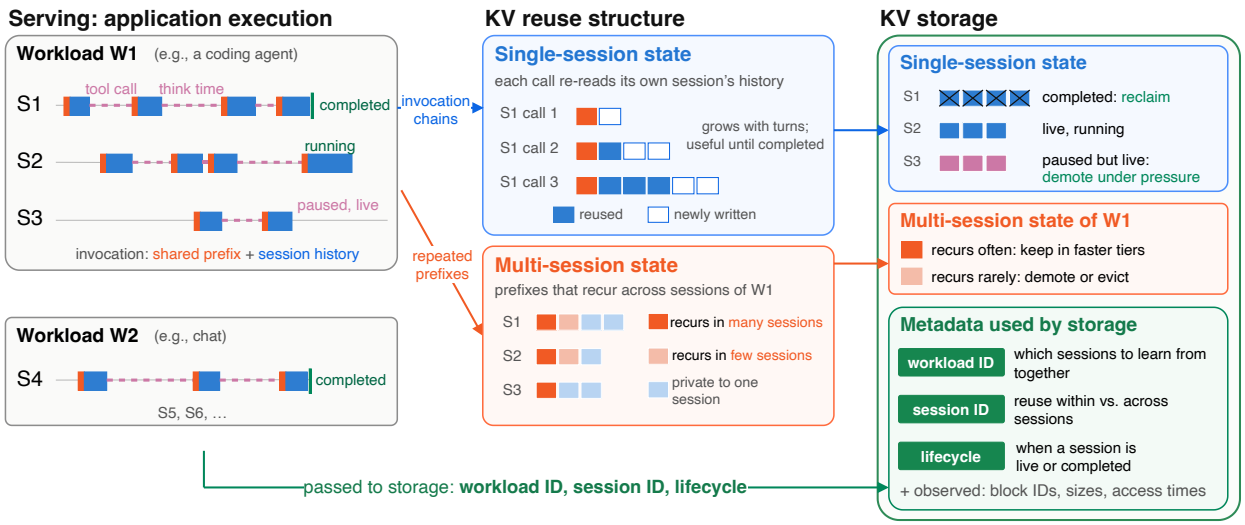


Figure 1: **Execution structure underlying KV-cache reuse.** Workloads contain multiple sessions, each comprising a sequence of model invocations. This structure produces *single-session state*, reused within one session, and *multi-session state*, reused across sessions. KV storage can infer this structure from a small amount of additional metadata.

These signals capture block accesses, but not the execution structure around them; for example, which accesses belong to one session, which sessions belong to the same recurring workload, or when a session has completed.

Prior systems demonstrate that exposing information from above the storage layer can improve KV management. They use application structure, workflow graphs, tool-call behavior, or agent-specific information to guide retention, prefetching, and eviction (Lin et al., 2024b; Li et al., 2025; Tiwari et al., 2026; Ren et al., 2025). This raises a broader question for a shared KV store serving heterogeneous applications: *how much information from above does storage actually need?* Can a general-purpose store adapt to workload behavior using a **small, generic interface** rather than understanding each application's semantics?

Existing workload studies do not directly answer this question. Request-level studies characterize arrivals and token counts (Wang et al., 2024; Patel et al., 2024); workload-specific studies characterize sessions, tool calls, and agent steps within individual application families (Zhu et al., 2026; Yuan et al., 2026); and KV-cache studies examine reuse within particular deployments (Wang et al., 2025a). What is missing is a **cross-workload view** of the KV blocks a shared storage layer actually manages: where their reuse comes from, when and how frequently they recur, how long they remain useful, and how their footprint evolves.

We characterize KV-cache reuse, lifetime, and footprint at the granularity of KV-cache blocks across 11 production and public traces, which contain 15 workloads (Qwen-Bailian and τ^2-bench each contribute three) in six categories: chat, API serving, RAG, coding agents, tool-using agents, and multi-lineage workflows. For traces containing request contents, we reconstruct the prefix-chained KV-block accesses visible to storage; we combine these with released block-level traces and production traces containing timing and execution metadata. Unless otherwise noted, we measure workload behavior without imposing a particular cache capacity or eviction policy.

Figure 1 illustrates the execution structure that motivates our analysis. A workload is a recurring serving context containing multiple sessions, with each session comprising a sequence of model invocations. At the storage layer, this hierarchy induces two distinct scopes of reuse: state reused within one session (*single-session state*) and state that recurs across sessions of the same workload (*multi-session state*). Our analysis reveals four main insights.

Reuse scope is critical when deciding which KV state to retain. As sessions deepen, reuse increasingly comes from repeatedly reading the session's own accumulated history, while shallow workloads can derive most of their reuse from content recurring across sessions, such as system prompts, templates, and retrieved documents. Consequently, similar aggregate hit ratios can represent different storage behavior: RAGBench and WildChat achieve similar potential hit ratios (71% and 74%), but the former is dominated by multi-session reuse while the latter primarily reuses single-session history. Branching further weakens the relationship between session length and reuse: Open Deep Research averages 39.5 model invocations (turns) per task but achieves only 20% potential reuse because those invocations are spread

across many short context lineages.

The distinction matters because the two forms of reuse imply different retention horizons and beneficiaries. A hit on single-session state indicates value tied to one live execution, so storage should manage that state with the session and reclaim it when the session ends. A hit on multi-session state demonstrates value beyond the current execution, so storage should evaluate it based on recurrence across sessions rather than the lifetime of any one session, using recurrence to decide whether it belongs in a fast tier or a colder one. Thus, aggregate hit rate alone is insufficient for retention and placement; *storage must know the scope of reuse*.

Idle time is a poor signal for when KV state should be discarded. Single-session state can remain useful through long periods with no model activity, including human think time and tool execution, but becomes dead when the session completes. Using an idle timer to approximate this boundary is difficult because the right timeout varies substantially across workloads. In Qwen To-C (a chat workload), a one-minute timeout retains only 22% of within-session hits and even a five-minute timeout loses 27%. Yet longer timeouts retain substantial dead state after sessions complete: with a five-minute timeout, completed-session state accounts for 51% of resident token-seconds in Qwen To-C, 61% in RAG-Pulse, and 40% in BrowseComp+. No single timeout avoids both premature eviction and unnecessary retention across these workloads. An explicit session-completion signal therefore gives storage a cleaner reclamation boundary for single-session state, while multi-session state must instead be managed according to its recurrence across sessions.

The source of capacity pressure determines the right storage response. Single-session state and multi-session state put pressure on cache capacity in different ways. Single-session state grows as sessions deepen and accumulates across concurrently live sessions. Stalls can leave much of this state idle but still live: with 100 sessions live at once, paused sessions account for 27–96% of resident single-session state across the production and agent traces we study. Multi-session state creates a different problem. Although it is stored only once across sessions, its shared pool contains a large low-frequency tail: in seven of the 13 workloads with block identities, 71–91% of multi-session memory belongs to blocks accessed by at most five sessions.

These two sources of pressure call for different controls, regardless of whether workloads share one KV store or are partitioned across separate storage pools. For single-session state, storage should track capacity by session, demote paused-but-live state when pressure rises, and reclaim it when the session completes. For multi-session state, storage should rank blocks by observed recurrence within the relevant workload or storage partition, keeping frequently reused blocks in faster tiers while demoting or evicting the low-frequency tail. The key distinction is therefore not whether different workload types compete directly for the same capacity, but that single-session pressure is driven by live-session footprint and concurrency, whereas multi-session pressure is driven by how broadly and frequently shared state recurs. Treating both populations with the same retention policy can either evict state still needed by live sessions or retain rarely reused shared state at the expense of more valuable KV state.

A small amount of execution context exposes substantial management opportunities. Storage already observes block identities, accesses, footprint, recurrence, and capacity pressure. Our findings suggest that much of the missing execution structure can be exposed through three generic signals from the serving layer: an opaque workload identifier, a session identifier, and session completion signal. We directly quantify what each signal adds beyond block-level observations alone. Scoping idle-time behavior by workload retains up to 22% more within-session hits or reduces dead state by up to 40% compared with a single global timeout. Session completion eliminates the idle-time tradeoff for single-session state, retaining useful state while a session is live and reclaiming it when the session ends. Finally, counting reuse across distinct sessions rather than across requests identifies shared state 4–700$\times$ more selectively, avoiding repeated private accesses being mistaken for cross-session sharing.

These measurements do not imply that the three signals are sufficient for an optimal KV-management policy, nor do they predict an end-to-end throughput or TTFT improvement. Rather, they show that generic execution context resolves ambiguities that materially change which KV state a storage policy would retain, reclaim, or demote. This provides a path toward workload-aware KV storage without requiring storage to interpret whether an application is chat, RAG, an agent, or another workload class.

In summary, this paper makes three contributions:

- We characterize KV-cache reuse, lifetime, and footprint at KV-block granularity across 11 production and public traces spanning six categories, using a common methodology for heterogeneous traces.
- We identify reuse scope as a key organizing principle: single-session and multi-session state exhibit different reuse, lifetime, reclamation, and capacity behavior.
- We quantify the value of generic execution context: workload identity scopes adaptation, session identity distinguishes private from shared reuse, and lifecycle identifies reclaimable single-session state, without requiring application-specific semantics.

2 BACKGROUND

2.1 KV Blocks and Multi-Tier Storage

KV blocks and prefix reuse. During LLM inference, the serving engine (e.g., vLLM (Kwon et al., 2023), SGLang (Zheng et al., 2023)) retains the key and value tensors computed for previously processed tokens so that later decoding steps and model invocations need not recompute them. KV-cache capacity grows with context length and model geometry and can consume a large fraction of GPU memory (Kwon et al., 2023; Sheng et al., 2023; Lin et al., 2024a). Modern serving engines therefore manage KV state in fixed-size *KV blocks*, each representing the state for a range of tokens across the model's layers (Kwon et al., 2023; Zheng et al., 2023; Liu et al., 2025). For example, vLLM's PagedAttention maps logical KV blocks to physical memory through a block table (Kwon et al., 2023). KV blocks are also the granularity at which state is identified, reused, transferred, and evicted, making them the natural unit for our analysis.

A cached block can be reused when a later invocation encounters the same token sequence following the same prefix. Prefix caching in these engines exploits this property to avoid redundant prefill computation. Reuse can occur within one execution as later turns revisit accumulated history, or across executions through recurring system prompts, templates, documents, or other shared prefixes (Gim et al., 2024; Qin et al., 2025; Wang et al., 2025a). Useful KV state can therefore outlive the invocation that created it.

Multi-tier KV storage. GPU memory cannot retain all potentially reusable KV state as contexts and concurrency grow. Systems therefore extend KV capacity into host DRAM, SSDs, and remote or disaggregated memory (Sheng et al., 2023; Lee et al., 2024; Gao et al., 2024; Yu et al., 2025; Chen et al., 2025a). LMCache (Liu et al., 2025), Mooncake (Qin et al., 2025), SGLang HiCache (Zheng et al., 2023), and related systems retain and move KV blocks across such tiers. Because the tiers differ in latency, bandwidth, and capacity, KV storage must decide which blocks to retain, when to reclaim them, and where to place them. Existing mechanisms primarily use storage-visible signals such as accesses, recency, occupancy, and TTL, which describe block behavior but not the execution structure that produced it.

2.2 Execution Context for KV Management

What storage sees. A KV storage layer naturally observes the objects it manages: block identities and sizes, reads and writes, access times, recurrence, occupancy, and pressure on its storage tiers. This information is sufficient to measure how often and how recently a block has been accessed, but it does not identify the execution boundaries around those accesses.

Three pieces of execution context are particularly relevant to our study. A *workload identifier* groups sessions belonging to the same recurring application or serving context and gives storage a scope over which to learn cross-session behavior. A *session identifier* groups accesses belonging to one logical execution and lets storage distinguish reuse within a session from reuse across sessions. Finally, *session lifecycle* indicates when an execution has completed, providing an explicit reclamation boundary for state whose value is confined to that session. These identifiers can be opaque to storage; they need not reveal application semantics.

> *Our study asks whether storage-visible block behavior, augmented with a small amount of generic workload and session context, can capture the distinctions that pertain to retention, reclamation, and placement.*

2.3 Related Work

KV-cache storage and management. A growing body of work extends KV capacity beyond GPU memory and develops mechanisms for retaining, transferring, and managing reusable KV state. CachedAttention (Gao et al., 2024) and Pensieve (Yu et al., 2025) target stateful multi-turn serving, while Mooncake (Qin et al., 2025), LMCache (Liu et al., 2025), and SGLang (Zheng et al., 2023) provide mechanisms for storing and moving KV state across memory and storage tiers. Other systems specialize management further; for example, IMPRESS (Chen et al., 2025a) uses importance information for multi-tier KV storage, and CacheGen (Liu et al., 2024) compresses KV state for transfer. Our work is complementary to these systems. Rather than proposing another KV-management mechanism, we characterize the reuse, lifetime, and capacity demand presented to such a storage layer across heterogeneous workloads, and ask what execution context is needed for making its management decisions.

Application- and workflow-aware KV management. Several recent systems exploit information above the KV-storage layer. Parrot (Lin et al., 2024b) exposes semantic variables and application dependencies, KVFlow (Pan et al., 2025) exploits multi-agent workflow structure, and PBKV (Zheng et al., 2026) predicts future invocations in dynamic agent workflows. MORI (Xia et al., 2026) exploits tool-call idle windows for offloading, and CacheWise (Tiwari et al., 2026) uses characteristics of coding-agent workloads to guide KV management. Continuum (Li et al., 2025) and AgServe (Ren et al., 2025) are related in making sessions explicit: Continuum retains a session's KV state across tool calls, and AgServe performs session-aware eviction.

These systems demonstrate that execution context can improve KV management, but generally target a particular application family or exploit application-, agent-, tool-, or workflow-specific structure. Our question is different: **across heterogeneous workloads, how much of the storage-relevant execution structure can be represented using generic signals?** Our characterization identifies workload identity, session identity, and session lifecycle as a small common interface that exposes the distinctions behind the storage behaviors we observe. We do not claim that richer information is unnecessary; it may improve prediction before reuse or future execution is observed.

LLM workload characterization. Prior studies characterize LLM serving from several complementary perspectives. BurstGPT (Wang et al., 2024) and Splitwise (Patel et al., 2024) characterize request arrivals, token distributions, and serving behavior, while KVCache in the Wild (Wang et al., 2025a) compares KV-cache behavior across several request types in production, including consumer-facing and API workloads. More recent work studies increasingly complex agentic applications. TraceLab (Zhu et al., 2026) and CacheWise (Tiwari et al., 2026) focus on coding agents, while AgentSysBench (Chang et al., 2026) compares ten agentic applications, including RAG, coding, research, and interactive agents, from an end-to-end serving perspective. Nixon et al. (Nixon et al., 2026) provide a complementary longitudinal view of production LLM serving across many users and models.

Our study provides a different cross-workload view: we **normalize** conventional and agentic workloads spanning chat, API serving, RAG, coding agents, tool-using agents, and multi-lineage workflows into a common KV-block representation. This lets us compare reuse scope, lifetime, recurrence, and capacity behavior at the granularity where a shared multi-tier KV store makes retention, reclamation, and placement decisions. Prior characterization has not combined this **breadth of workload structures** with a **common storage-level analysis of KV state**.

3 Methodology and Workloads

We characterize KV behavior from the perspective of a shared, multi-tier storage layer. We normalize each trace into a stream of KV-block creations and accesses, together with timestamps and execution metadata when available.

3.1 Representation and Reuse Scope

A *workload* is a recurring serving context containing multiple *sessions*; a session is one logical execution containing one or more model invocations (turns). Workload and session IDs can be opaque.

We classify reused KV state by its scope. A block accessed only within one session is *single-session state*, while a block accessed by multiple sessions of the same workload is *multi-session state*. Correspondingly, repeated accesses from the same session are *within-session hits*, and accesses from another session are *cross-session hits*. Given workload and session IDs, storage can distinguish these hit scopes online.

Block classification is retrospective: a block observed in one session may later become multi-session state. We therefore classify blocks using their complete observed trace, while the storage implications we derive use only information available at each access. Richer annotations such as semantic roles and tool types are used only to explain behavior; workload categories such as chat or RAG are descriptive and are not workload identifiers.

3.2 Workloads

We analyze 11 production and public traces, which contain 15 workloads in six categories. Qwen-Bailian and τ^2-bench each contribute three workloads, which we analyze separately because they show distinct KV-access behavior. Table 1 in Appendix A.1 summarizes the source, scale, and fields available for each workload. Open-ended **chat** includes WildChat (Zhao et al., 2024), BurstGPT (Wang et al., 2024), and Qwen-Bailian To-C (Wang et al., 2025a); **API serving** uses Qwen-Bailian To-B (Wang et al., 2025a); **RAG** includes RAGBench (Friel et al., 2024) and RAG-Pulse (Wang et al., 2025b); **coding agents** include Exgentic SWE-bench (Bandel et al., 2026; Jimenez et al., 2024), Qwen-Bailian Code (Wang et al., 2025a), and TraceLab (Zhu et al., 2026); and **tool-using agents** include AppWorld (Trivedi et al., 2024), BrowseComp+ (Chen et al., 2025b), and the three τ^2-bench domains (Yao et al., 2024b) from Exgentic (Bandel et al., 2026).

Our **multi-lineage** workload is Open Deep Research - ODR (LangChain AI, 2025), a LangGraph application in which a supervisor delegates work to concurrent sub-researchers, creating multiple independently growing context lineages within one session. Because no public trace exposes the model inputs needed to reconstruct this behavior, we collect a trace by running 50 English tasks from DeepResearch Bench (Du et al., 2025). We serve Qwen3-8B (Qwen Team, 2025) with vLLM (Kwon et al., 2023) on one NVIDIA A100 GPU and record every model request and response. Appendix A.1 gives further collection details.

3.3 Trace Normalization

Traces differ in model, tokenizer, and block size, which change how requests map to KV blocks and bytes; we normalize them to a common block-level representation so that differences reflect workloads, not deployments. Blocks generated during decode are stored and become reusable from

the session's next call; outputs of a session's final call are never read and count only toward footprint. Each analysis uses only workloads that provide the required fields.

Raw-text traces. WildChat, RAGBench, the Exgentic workloads, and Open Deep Research provide model inputs and session structure. We tokenize requests using the Qwen3-8B tokenizer (Qwen Team, 2025), divide them into 16-token blocks, and compute vLLM-style prefix-chained hashes. Blocks match only when they contain the same tokens following the same prefix. Because the tokenizer may differ from that used in the original deployment, these traces are normalized replays of request structure rather than exact reconstructions of the original KV layout. Appendix A.2 evaluates sensitivity to block size.

Pre-hashed traces. Qwen-Bailian and RAGPulse provide block identifiers directly. Qwen-Bailian uses 16-token prefix-chained blocks and includes session linkage. RAGPulse identifies 512-token content chunks independently of prefix position, so we chain its identifiers along each prompt to obtain prefix-exact identities. Its larger blocks hide shorter shared prefixes, so we use RAGPulse primarily for timing and lifetime analyses rather than direct reuse comparisons.

Metadata-only traces. BurstGPT and the production timing portion of TraceLab provide timestamps, token counts, and session structure but not reconstructable block identities. We use these data only for idle-time, concurrency, and projected footprint analyses; Appendix A.1 describes how we use them in detail.

4 Cross-Workload Analysis

We examine three properties of KV state that affect multi-tier storage: reuse scope (§4.1), lifetime (§4.2), and capacity demand (§4.3). Unless noted otherwise, measurements assume unlimited capacity and no eviction, so they characterize behavior in the access stream rather than behavior induced by a particular cache policy.

4.1 Reuse Scope and Concentration

Finding 1: *Reuse scope determines how storage should manage reused KV state.*

Substantial KV reuse occurs across workload types, but its source differs sharply. We define *potential hit ratio* as the fraction of block reads that would find the block already present under unlimited capacity. Among the representative workloads, it ranges from 20% in Open Deep Research (ODR) to 99% in τ^2-Telecom (Figure 2a). Reuse is also concentrated: at most 40% of unique blocks supply 80% of hits in every workload shown, and Qwen To-B requires only

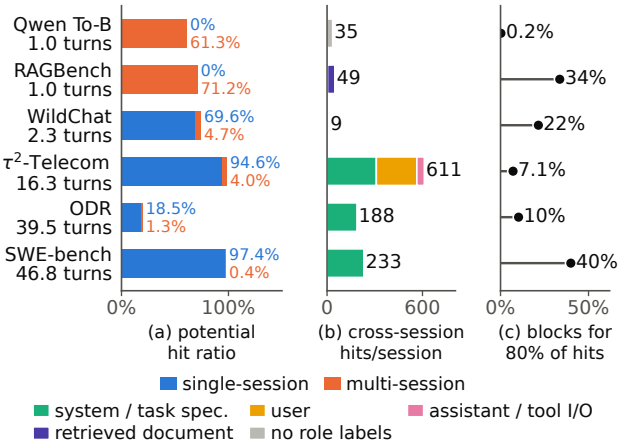


Figure 2: Reuse in six representative workloads, ordered by turns per session. (a) Potential hits split into within-session and cross-session reuse. (b) Cross-session hits per session, broken down by semantic role where available. (c) Fraction of unique blocks supplying 80% of hits.

0.2% (Figure 2c).

Subfinding 1.1: *Session structure determines the source of reuse; aggregate hit ratios hide this distinction.*

Each turn in a linear session rereads context accumulated by earlier turns, so deeper sessions naturally create more within-session reuse. Single-turn Qwen To-B and RAGBench obtain all hits across session boundaries, whereas WildChat, with 2.3 turns per session, already obtains most reuse from its own history. τ^2-Telecom and SWE-bench, with mean 16.3 and 46.8 turns per session, respectively, reach 99% and 98% potential hit ratios dominated by within-session reuse.

Similar aggregate hit ratios can therefore represent different storage behavior. RAGBench and WildChat achieve 71% and 74% potential hit ratios, but RAGBench is dominated by cross-session reuse while WildChat primarily rereads single-session history.

Branching breaks a simple relationship between task length and reuse. ODR averages 39.5 model invocations per task but only 20% potential reuse because those invocations are distributed across roughly 33 independently evolving context lineages, each averaging 1.2 turns. The relevant quantity is therefore the depth of a prefix lineage rather than the total number of invocations in the task.

Cross-session fractions can also hide substantial absolute reuse (Figure 2b). Only 0.4% of SWE-bench hits cross sessions, yet each session has 233 cross-session hits, compared with 9 in WildChat; 99% are on the system prompt and tool definitions shared across independent coding problems. τ^2-Telecom has 611 cross-session hits per session.

Deep workloads can therefore contain substantial single- and multi-session reuse simultaneously, even when the latter is a small fraction of all hits.

Storage implication. Reuse scope should guide retention and placement. Single-session reuse is valuable only while its session remains live, whereas cross-session reuse provides evidence of value beyond the current execution. Aggregate hit rate alone therefore does not reveal how long or where reused state should be retained.

Subfinding 1.2: *Prefix recurrence, not semantic role, determines sharing.*

Cross-session reuse requires identical content to recur after an identical prefix. Stable system prompts often satisfy this condition, but other semantic regions can as well. Using trace annotations, we group blocks into system or task-specification, user, assistant or tool-I/O, and retrieved-document regions. In τ^2-Telecom, user blocks contribute roughly two-fifths of its 611 cross-session hits per session, with assistant and tool blocks contributing substantial additional reuse (Figure 2b).

Loaded content illustrates the same principle. RAGBench repeatedly draws from a fixed document corpus and obtains 46 of its 49 cross-session hits per session from retrieved content, whereas ODR issues diverse web queries and obtains essentially no cross-session reuse from retrieved pages. Large retrieved inputs therefore do not imply reusable KV state: content must recur under a stable prefix. ODR also places the same research brief after different prefixes in different lineages, preventing reuse.

Storage implication. Storage need not understand semantic roles to recognize realized sharing. Observed recurrence across sessions provides a generic signal for identifying and ranking multi-session state; semantic hints may still help predict reuse before it occurs.

4.2 Lifetime and Recurrence

Finding 2: *Idle time is a poor signal for when KV state should be discarded.*

Single-session state can remain useful while its session is idle, but its value ends when that session completes. Without lifecycle information, storage must approximate this boundary using an idle timeout, which can fail in both directions.

Subfinding 2.1: *Setting time to live is challenging for both single-session and multi-session state.*

The appropriate timeout varies substantially across workloads (Figure 3a, b). In Qwen To-C, a one-minute timer retains only 22% of within-session hits and a five-minute timer

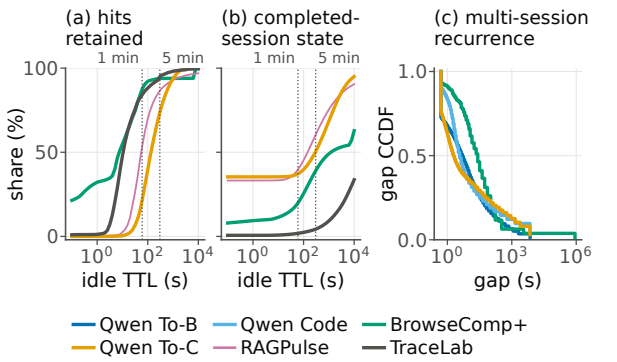


Figure 3: Lifetime of single- and multi-session state. Using an idle timer to approximate single-session lifetime: (a) within-session hits retained and (b) share of resident token-seconds held by sessions that have already completed, as the timer grows; (c) recurrence gaps for multi-session state; tails are bounded by collection window (Table 1).

still retains only 73%. BrowseComp+ retains 88% with a one-minute timer, while TraceLab retains 48% with a ten-second timer. Longer timers reduce premature expiration but retain dead state: with a five-minute timer, completed-session state accounts for 51% of resident token-seconds in Qwen To-C, 61% in RAGPulse, and 40% in BrowseComp+, compared with 4% in TraceLab. No single timeout avoids both errors across these workloads.

When available, session completion removes this ambiguity for single-session state by providing an explicit reclamation boundary. Some workloads, particularly open-ended chat, may not expose a definitive completion event. In such cases, idle time can still guide demotion to a lower tier, but cannot reliably distinguish a paused session from one whose state is safe to reclaim.

Multi-session state has no session-completion boundary because completing one session does not invalidate state that may recur in another. Instead, its recurrence gaps span several orders of magnitude (Figure 3c). In the four workloads shown, 69–80% of gaps are under a minute, yet 11–18% exceed five minutes and 4–10% exceed an hour. Counted in intervening sessions, the spread is wider still: BrowseComp+'s common prefix recurs almost every session, whereas half of Qwen To-B's gaps exceed 100 sessions (Appendix A.4). Thus, multi-session state ranges from continuously recurring prefixes to blocks unused for hours.

Storage implication. The two reuse scopes need different lifetime signals. Session completion, when available, gives single-session state an explicit reclamation boundary; otherwise idle time can guide demotion but not definitive reclamation. Multi-session state should instead be ranked

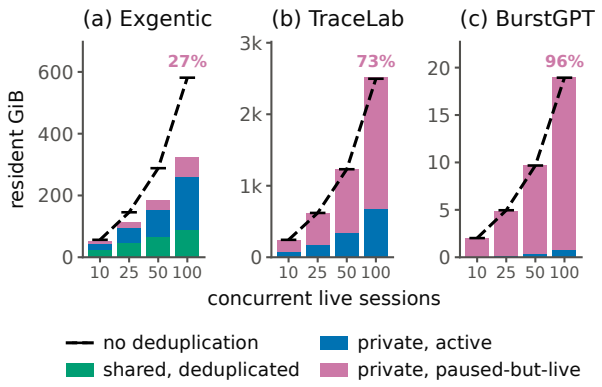


Figure 4: Resident KV state under increasing concurrency, split into multi-session state and active versus paused single-session state. Percentages report the paused share of single-session state at 100 live sessions. Exgentic pools its six agent workloads.

by recurrence across sessions.

Subfinding 2.2: *Stalls create idle-but-live state whose cost grows with overlapping demand.*

Tool execution, API latency, and human think time can leave a live session inactive even though later invocations still require its KV state. This paused-but-live state consumes fast-tier capacity, but reclaiming it would force a later reload or recomputation; the pressure grows as more live sessions overlap.

With 100 sessions live at once, paused sessions account for 27% of single-session resident state in Exgentic, 73% in TraceLab, and 96% in BurstGPT (Figure 4; setup in Appendix A.5). The absolute footprint depends on how much state each session has accumulated: TraceLab's long coding sessions create substantially more resident KV state than the short conversational sessions in BurstGPT at the same concurrency.

Storage implication. Inactivity should not be treated as death. Under capacity pressure, storage can demote paused-but-live state while session lifecycle distinguishes it from state that can be reclaimed.

4.3 Footprint and Capacity

Finding 3: *The source of capacity pressure determines the right storage response.*

Single-session and multi-session state consume capacity differently. Single-session state grows with each live execution and is replicated across concurrent sessions. Multi-session state is stored once across sessions, but can accumulate a large tail of blocks that recur infrequently. For a common byte scale, we normalize tokens using the KV geometry of Qwen3-8B in `bf16` (144 KiB per token). Appendix A.3 reports the corresponding measurements for all workloads.

Subfinding 3.1: *Session footprint depends on both initial context and growth.*

Figure 5a shows how single-session footprint evolves with session depth. Across multi-turn workloads, median context grows approximately linearly with session depth ($R^2 \geq 0.95$), but both starting footprint, determined by initial context such as system prompts, tool definitions, and loaded content, and subsequent growth rate vary widely. AppWorld begins at 12.1 GiB and τ^2-Telecom at 3.1 GiB, whereas WildChat and Qwen To-C begin at 0.01 and 0.11 GiB. Per-turn growth varies by roughly $20\times$, from 11 MiB in τ^2-Telecom to 222 MiB in BrowseComp+. These two quantities can reverse which workload dominates capacity: BrowseComp+ begins at roughly half the footprint of τ^2-Telecom but, because it grows about $20\times$ faster per turn, surpasses it within ten turns.

Session depth determines how far this growth accumulates. The median τ^2-Telecom session lasts 13 turns but its 99th percentile reaches 105; TraceLab has a median of 17 turns but a 99th percentile above 1,000. Neither initial context size nor typical session length alone therefore captures capacity demand.

Storage implication. Single-session capacity should be accounted for per live session rather than through an application-wide fixed reservation. Storage can estimate demand from each session's observed footprint and growth, combined with the number of concurrently live sessions.

Subfinding 3.2: *Most multi-session KV volume belongs to blocks reused by few sessions.*

Figure 5b compares how much memory is required to capture within-session and cross-session reuse. The private state of only six concurrent SWE-bench sessions already exceeds the 17.6 GiB required to supply 90% of its cross-session hits.

The multi-session pool contains a pronounced low-frequency tail. Frequently recurring prefixes form a compact hot head: half of cross-session hits require only 0.09 GiB in Qwen To-C and 2.4 GiB in Qwen To-B. The complete Qwen To-B multi-session pool, however, reaches 479 GiB. RAG-Bench is less concentrated, requiring 443 GiB to supply half of its cross-session hits.

Figure 5c shows that much of the multi-session pool is reused by only a few sessions. In five of the six workloads shown, blocks accessed by at most five sessions account for 71–91% of multi-session memory; Appendix A.3 re-

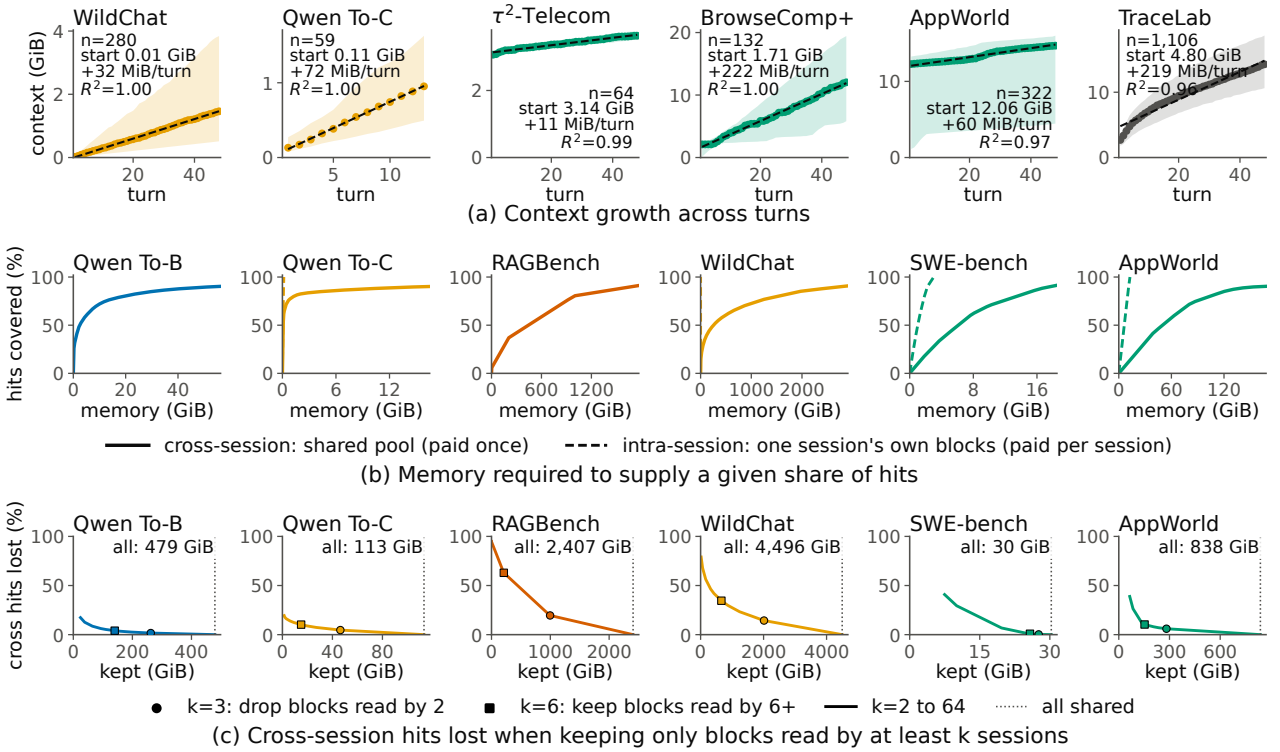


Figure 5: KV footprint across representative workloads. (a) Context growth with session depth. (b) Memory needed to supply a given fraction of cross-session hits. (c) Multi-session memory retained and hits lost when filtering blocks by the number of sessions that reuse them.

ports all workloads. Their usefulness nevertheless differs across workloads: discarding these blocks loses only 4% of cross-session hits in Qwen To-B but 63% in RAGBench. A fixed reuse-frequency threshold is therefore unlikely to work across workloads.

Storage implication. Multi-session state should not be treated as uniformly valuable. Observed cross-session recurrence can rank the shared pool, keeping frequently recurring blocks in faster tiers while demoting or evicting the low-frequency tail.

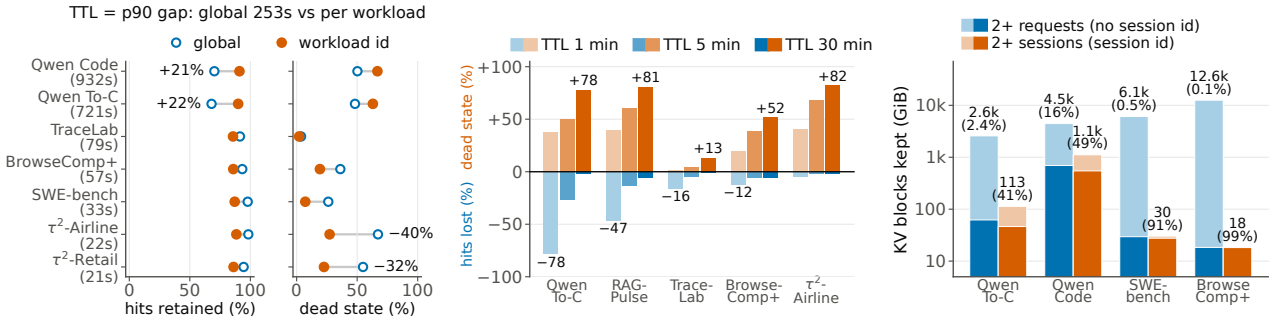
4.4 Application-Independent Execution Context

Across these findings, **storage decisions depend on reuse scope, lifecycle, recurrence, and capacity cost rather than application type**. Most of this information is already visible to storage: block accesses reveal reuse and recurrence, allocations reveal footprint, and occupancy exposes capacity pressure. The missing information is execution structure: which sessions belong to the same workload, which accesses belong to one session, and when that session completes.

Figure 6 quantifies what each of the three execution signals adds beyond block-level observations alone.

Workload ID. Storage can learn a timeout value across all the sessions it serves; however, with workload identity, storage can adapt the timeout value to each workload to be significantly more accurate. Figure 6a illustrates this difference by setting the TTL to the 90th percentile of observed reuse. The pooled TTL of 253 s is too short for Qwen To-C and Qwen Code: per-workload estimates retain 22% and 21% more within-session hits. Conversely, per-workload estimates reduce dead state (state that can be safely dropped but is retained) by 40% in τ^2-Airline and 32% in τ^2-Retail. The workload ID does not improve both metrics for every workload; it lets each workload operate a timeout suited to its own reuse distribution.

Session lifecycle. Figure 6b compares idle timeouts with reclamation at session completion. Completion provides an explicit end-of-life signal for single-session state: it preserves all within-session hits while immediately reclaiming state that can no longer be reused. No fixed timeout achieves both. A one-minute TTL loses 78% of within-session hits in Qwen To-C and 47% in RAGPulse; a 30-minute TTL keeps nearly all of them, but dead state then makes up 78–82% of



(a) Workload ID (b) Session lifecycle (c) Session ID

Figure 6: What each piece of execution context adds. (a) Workload ID: idle TTL set to the 90th percentile of observed within-session gaps, estimated over all seven workloads or over each workload's own sessions. (b) Session lifecycle: idle TTL of 1, 5, and 30 minutes compared with reclaiming state at session completion. (c) Session ID: blocks flagged as reused at their second access, counted per request (no session ID) or per distinct session (session ID), with causal counts and no capacity limit. Each full bar is the state kept as reused; its dark part is the state that a later session actually reads, with that share in parentheses.

held state in Qwen To-C, RAGPulse, and τ^2-Airline, and 52% in BrowseComp+. The completion signal removes this trade-off for single-session state.

Completion alone is insufficient for handling multi-session state: in Qwen Code, where 53% of re-reads cross sessions, reclaiming every block no longer held by a live session retains 37% fewer hits than a five-minute TTL (Appendix A.6). Lifecycle-based reclamation therefore requires session identity to distinguish single-session from shared state.

Session ID. Without session identity, repeated reads within one execution can be mistaken for sharing across executions. Figure 6c flags a block at its second access, counted either per request or per distinct session.

Counting requests flags 2.6k–12.6k GiB in the four workloads, of which only 0.1–16% is later read by another session. Using session IDs flags 4–700× less state (18 GiB–1.1k GiB), and 41–99% of it is read by a later session. The session ID thus identifies the shared state that completion should spare, so lifecycle can reclaim single-session state without discarding blocks that later sessions read (Appendix A.6).

Together, the three signals resolve different ambiguities: workload ID scopes adaptation, session ID distinguishes private reuse from sharing, and session completion exposes liveness. Figure 6 shows that each materially changes which state a policy would retain or reclaim without requiring application-specific semantics.

5 Discussion

Scope of the characterization. Our trace-driven analysis characterizes demand presented to KV storage rather than a complete serving stack, and does not model transfer costs, scheduling, or a particular finite-capacity policy. Some statistics are measured offline rather than learned incrementally. Our results quantify the information exposed by workload, session, and lifecycle context, but do not predict the end-to-end performance of a specific storage policy.

Execution boundaries and trace normalization. We assume the serving layer can associate requests with workloads and sessions and, when available, expose session completion. Raw-text traces are normalized using one tokenizer, 16-token blocks, and the KV geometry of Qwen3-8B; Appendix A.2 shows that the main reuse and recurrence trends are stable across block sizes.

Complementary techniques. We study prefix-exact reuse within each workload. Reuse across differing prefixes (Yao et al., 2024a), attention-based KV selection (Zhang et al., 2023; Li et al., 2024), and cross-workload reuse are complementary to our work.

6 Conclusion

We present a storage-oriented characterization of KV-cache behavior across heterogeneous LLM workloads. We show that three pieces of generic execution context can materially improve how KV storage manages state. These results motivate workload-aware KV storage built on a small cross-layer interface rather than application-specific semantics.

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A Appendix

A.1 Trace Collection

Table 1 summarizes the traces used in this paper, including their workload type, source, scale, and timestamp origin. *Sess.* and *Reqs.* report the number of sessions and model invocations, *Turns* is the mean number of invocations per session, and *Input* is the mean prompt length in tokens.

The Exgentic traces combine runs from four to five models; their timestamps come from agent execution and therefore include real tool latency. RAGPulse uses 512-token blocks rather than the 16-token blocks used by the other block-level traces; its byte-based volume measurements remain comparable, but its reuse measurements can hide shared prefixes shorter than one block. RAGPulse reconstructs its context independently at each turn and therefore has no cumulative session growth.

Metadata-only traces. TraceLab and BurstGPT contain no prompt text or block identities. TraceLab provides, for each model call, its session, its position in the session (round index), start and end times, and input and output token counts. BurstGPT provides the same for the release window that includes session identifiers, except that it has no turn index, so we order a session's calls by arrival time; calls in a session almost never overlap and never share a start time. We therefore use these traces only for results that depend on timing and context size, and project KV state from token counts.

Where an analysis needs the state a call reuses from its own session, we take it to be the previous call's context, $\min(input_{k+1}, input_k + output_k)$ tokens, since chat and coding agents resend the conversation with each call unless they compact it. TraceLab also reports cached-prefix tokens, but these are the providers' prompt-cache hits and depend on the providers' caching policies, so they cannot serve as a policy-independent measure; for consecutive calls less than a minute apart (88% of calls), they agree with our estimate within 2%.

Open Deep Research (ODR) collection. We collected the ODR trace ourselves. Open Deep Research (LangChain AI, 2025) is a LangGraph application in which a supervisor writes a research brief, delegates subtopics to parallel researcher agents, and then produces a final report from their compressed findings. We run it on 50 English tasks (IDs 51–100) from DeepResearch Bench (Du et al., 2025), one task at a time, with user clarification disabled. All model roles use Qwen3-8B (Qwen Team, 2025) in bf16, served by vLLM (Kwon et al., 2023) on one NVIDIA A100 GPU with a 32,768-token context window, while web search uses Tavily. To fit within this context window, we reduce several upstream limits, including the number of concurrent researchers, supervisor rounds, tool-calling rounds, and output lengths. Our goal is not to reproduce the benchmark configuration or evaluate task quality, but to study the resulting KV-access patterns. These changes therefore affect the scale of individual executions rather than the role of the trace in our analysis of multi-lineage context and reuse behavior.

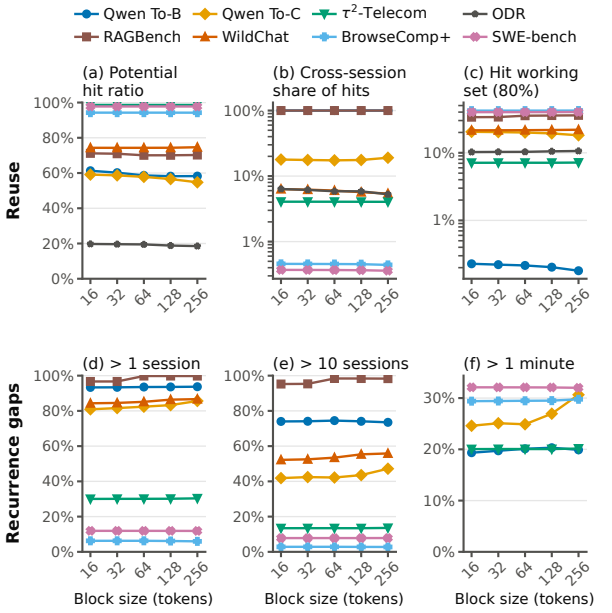

Figure 7: Reuse (top) and cross-session recurrence (bottom) for block sizes of 16–256 tokens.

We record every model call at the client's HTTP transport, capturing the exact request messages, response, and wall-clock start and end times of each chat-completion call, together with its task identifier. Because tasks run sequentially, the timestamps capture concurrency among researchers within a task but not across concurrent users. The trace contains 1,973 model invocations over 5.6 hours. Unlike a chat session, a single task contains many independent message histories: the supervisor, each researcher, and the report writer each maintain their own. We recover these histories, which we call *lineages*, from the messages themselves. A call continues a lineage if the messages and response of an earlier call appear unchanged at the start of its input; when multiple candidates match, we link it to the longest one. A call with no match starts a new lineage, such as a newly launched researcher. Each task contains about 33 lineages (§4.1). We then tokenize and block the calls using the same procedure as for the other raw-text traces (§3.3).

Trace	Workload type	Source	Sess.	Reqs.	Turns	Input	Timestamps
Raw text *Prompts and responses; 16-token blocks reconstructed with the Qwen3-8B tokenizer; semantic-role labels*							
WildChat (Zhao et al., 2024)	Open-ended chat	Production	838K	1.96M	2.3	1.4K	Real, 13 months
AppWorld* (Trivedi et al., 2024)	Tool-using agent	Benchmark	1,500	48K	32.3	83K	Agent run
BrowseComp+* (Chen et al., 2025b)	Tool-using agent	Benchmark	1,948	38K	19.7	44K	Agent run
SWE-bench* (Jimenez et al., 2024)	Coding agent	Benchmark	1,959	92K	46.8	22K	Agent run
τ^2-bench* (Yao et al., 2024b)							
Airline	Tool-using agent	Benchmark	957	11K	11.4	11K	Agent run
Retail	Tool-using agent	Benchmark	1,848	22K	11.9	11K	Agent run
Telecom	Tool-using agent	Benchmark	1,844	30K	16.3	15K	Agent run
RAGBench (Friel et al., 2024)	RAG	Benchmark	67K	67K	1.0	1.1K	Synthetic
Open Deep Research (LangChain AI, 2025)	Multi-lineage	Ours	50	1,973	39.5	6.1K	Agent run, 5.6 h
Pre-hashed *Released block identifiers, 16-token (Qwen-Bailian) or 512-token (RAGPulse); no text*							
Qwen-Bailian (Wang et al., 2025a)							
To-C	Open-ended chat	Production	23K	43K	1.9	2.3K	Real, 2 h
To-B	API serving	Production	173K	173K	1.0	0.9K	Real, 2 h
Code	Coding agent	Production	26K	43K	1.6	5.7K	Real, 2 h
RAGPulse (Wang et al., 2025b)	RAG	Production	19K	35K	1.9	3.6K	Real, 42 days
Metadata only *Timestamps, token counts, and session structure; no block identities*							
TraceLab (Zhu et al., 2026)	Coding agent	Production	4,281	357K	83.4	154K	Real, 8 months
BurstGPT v2 (Wang et al., 2024)	Open-ended chat	Production	55K	232K	4.2	1.0K	Real, 110 days

Table 1: The 11 traces and their 15 workloads, grouped by evidence class (§3.3). *Agent runs released by Exgentic (Bandel et al., 2026).

A.2 Block Sensitivity

All results in §4 use 16-token blocks. To test whether the findings depend on this choice, we repeat the reuse and temporal measurements with 32-, 64-, 128-, and 256-token blocks on eight workloads, including Qwen To-C and BrowseComp+, which anchor the temporal results. Because block identifiers are prefix-chained, a b-token block is identified by the last of its $b/16$ 16-token sub-blocks, so we derive the larger block sizes from the 16-token traces rather than re-tokenizing.

Reuse. Block size does not change the reuse picture (Figure 7a–c). The ordering of workloads is preserved in every panel, except that Open Deep Research and WildChat, whose cross-session shares are both 6.3% at 16 tokens, trade places. The agent workloads and WildChat keep the same hit ratio at every block size. Hit ratios decline slightly only in Qwen To-C (59% to 55% at 256 tokens), Qwen To-B (61% to 58%), RAGBench (71% to 70%), and Open Deep Research (19.7% to 18.5%). Cross-session shares move by at most 1.1 percentage points and hit working sets by at most 2.2 points. Larger blocks mainly reduce reuse near prefix boundaries: when two sequences diverge within a block, the shared tokens in that partially matching block can no longer be reused. This matters for short shared prefixes, but has little effect when reuse spans thousands of tokens, as in the agent workloads.

Recurrence. Recurrence gaps are equally stable (Figure 7d–f). In the agent workloads, every share stays within 0.6 percentage points. The short-context workloads move by at most 6 points, mostly upward. RAGBench shows why: every request begins with the same 32–47-token instruction prefix, which fills two 16-token blocks and one 32-token block but no larger block. From 64 tokens onward, this block, which every request shares, disappears, and the share of gaps longer than one session rises from 97% to nearly 100%. Qwen To-C and WildChat drift more gradually, consistent with the same loss of short shared prefixes: Qwen To-C's share of gaps above ten sessions rises from 42% to 47% and above one minute from 25% to 31%.

Context-type composition. The semantic role of a block is where block size matters most (Figure 8). Roles change at turn and message boundaries, so larger blocks increasingly span more than one context type, and their share grows with every doubling of block size. At 16 tokens, such blocks are 4–11% of unique blocks and supply at most 6% of hits, so the role attribution in §4.1 is reliable. At 256 tokens, they are 40% of unique blocks in WildChat, 52% in RAGBench, and 75% in τ^2-Telecom. Their share of hits stays lower in RAGBench and τ^2-Telecom, 25% and 20% at 256 tokens, because much of the reuse comes from long single-type runs such as retrieved documents and tool definitions; in WildChat, it reaches 46%. A role label per block is therefore a relatively reliable admission hint for small blocks but becomes ambiguous for large ones.

	Turns		Growth			Single-session state (GiB)		Multi-session state (GiB)		Multi-session tail (%)		
Workload	p50	p99	**Start (GiB)**	**MiB/ turn**	R^2	**Peak p99**	**Reused p50**	**All**	**50% of hits**	**Mem. k=2**	**Mem. $k\leq5$**	**Hits lost**
Single-request												
Qwen To-B	1	1	–	–	–	0.95	–	479	2.43	45	71	4
RAGBench	1	1	–	–	–	1.91	–	2,407	443	59	91	63
Short multi-turn												
Qwen Code	1	9	0.58	79	0.99	2.49	0.40	1,114	2.86	51	86	13
Qwen To-C	1	8	0.11	72	1.00	2.06	0.16	113	0.09	59	87	10
RAGPulse	1	9	–	–	–	1.46	0.14	30.0	0.14	77	80	1
WildChat	1	14	0.01	32	1.00	1.43	0.07	4,496	254	55	85	35
Agents												
τ^2-Airline	10	50	0.67	78	0.98	4.87	1.35	35.6	8.68	6	21	2
τ^2-Retail	11	51	2.60	28	0.95	4.35	1.48	36.5	8.27	10	23	1
τ^2-Telecom	13	105	3.14	11	0.99	4.41	1.55	140	12.4	31	60	6
BrowseComp+	13	96	1.71	222	1.00	24.7	4.62	18.4	5.18	1	9	0
AppWorld	19	150	12.1	60	0.97	31.2	12.7	838	49.9	66	82	10
SWE-bench	38	159	0.67	71	0.99	13.1	3.02	30.3	6.10	9	15	1
TraceLab	17	1,162	4.80	219	0.96	58.6	–	–	–	–	–	–

Table 2: Per-workload volume and footprint under the Qwen3-8B bf16 KV geometry (144 KiB per token).

Volume. We do not repeat the volume analysis: KV bytes are proportional to tokens, and the context-growth fits of Figure 5 are computed from token counts, so block size affects shared volume only through the short-prefix effect described above.

Storage implication. Block size does not change the conclusions of §4. It matters in two places. Shared prefixes shorter than a block are invisible to storage, so blocks used for cross-session sharing should be no larger than the shortest stable prefix a workload repeats. Role labels lose precision as blocks grow, so role-based hints need small blocks or token-range metadata. Deep private session state can use large blocks, which cut the number of block entries by 16× from 16 to 256 tokens without losing reuse.

A.3 Volume Summary

Table 2 reports the volume metrics of §4.3, grouped into single-request workloads, short multi-turn workloads, and agents. *Turns* reports the median and 99th-percentile number of invocations per session. *Growth* fits a least-squares line to the median context size at each turn among sessions that reach the same depth: *Start* is the intercept, *MiB/turn* is the slope, and R^2 reports goodness of fit.

For single-session state, *Peak* is the 99th percentile of each session's maximum context size, and *Reused* is the median amount of that session's own KV state needed to supply all of its intra-session hits. Because this state is private to a session, each concurrently live session incurs this footprint separately.

For multi-session state, *All* is the total size of blocks reused by more than one session. *50% of hits* reports the amount of this state needed to supply half of all cross-session hits when blocks are ranked by reuse frequency. The *tail* columns characterize low-frequency shared state: they report the fraction of multi-session memory occupied by blocks used by exactly two sessions (k=2) or by at most five sessions ($k\leq5$), together with the fraction of cross-session hits lost if blocks used by at most five sessions are discarded.

A.4 Recurrence Gaps in Sessions

Figure 9 measures the gaps of Figure 3c in intervening sessions rather than seconds. RAGBench has no real timestamps, so it is absent from Figure 3c but included here. BrowseComp+'s common prefix recurs almost every session: only 6% of gaps exceed one session. In Qwen To-B, half of the gaps exceed 100 sessions and 9% exceed 10,000; in RAGBench, 79% exceed 100 sessions and 45% exceed 1,000.

A.5 Fixed-Concurrency Replay

Figure 4 keeps exactly N sessions live at all times. Each of N slots replays randomly chosen recorded sessions from the trace back to back, starting a new one as soon as the previous one ends. Each session keeps its recorded call times and context sizes. A session is live from the start of its first call to the end of its last. We convert tokens to bytes at 144 KiB per token, the bf16 KV size of Qwen3-8B. The Exgentic panel draws sessions from all six Exgentic

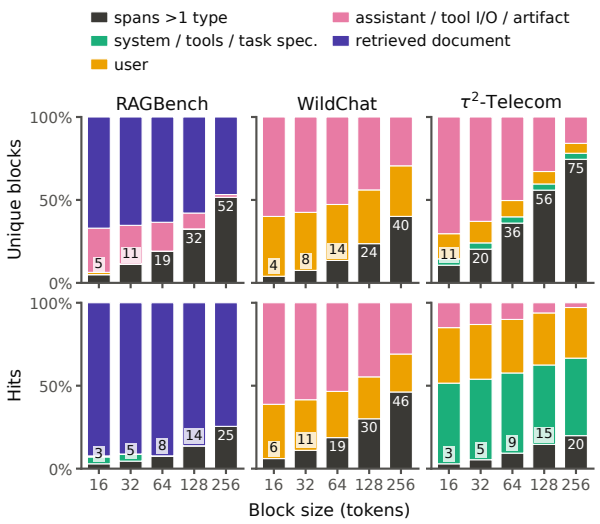

Figure 8: Context-type composition of unique blocks (top) and hits (bottom) for block sizes of 16–256 tokens. Black: blocks whose tokens span more than one context type; the number gives their share in percent.

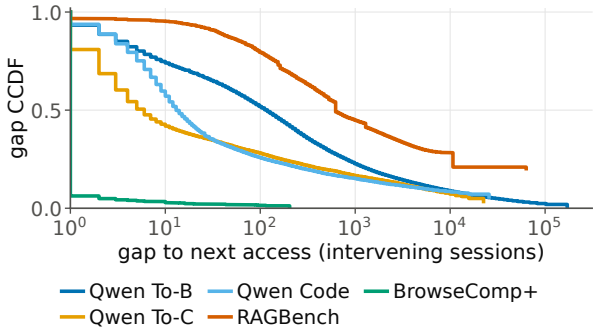

Figure 9: Recurrence gaps of multi-session state, counted in session start order: a gap of one means the next session to start reuses the block.

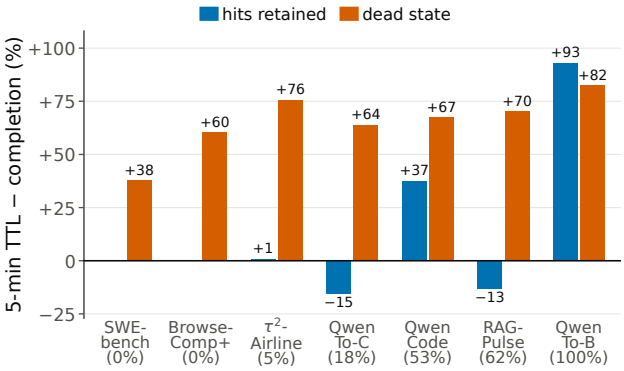

Figure 10: A five-minute idle TTL minus reclamation at session completion, for all hits retained (within- and cross-session) and dead state. Brackets: share of re-reads that are cross-session.

workloads (AppWorld, BrowseComp+, SWE-bench, and the three τ^2-bench domains). In Exgentic, blocks shared with other sessions are counted once while any session that uses them is live. BurstGPT and TraceLab expose no block identities, so all of their state is counted as private. The dashed line instead counts each live session's full context. Each bar reports the mean resident state once the replay reaches a steady state.

A.6 Session Completion and Shared State

Figure 6b considers only within-session hits, for which reclamation at completion is lossless. Figure 10 extends the replay to include cross-session reuse. At a session's final request, completion frees every block touched by that session unless another live session still holds it; the TTL instead frees blocks after five minutes of inactivity. Dead state is the share of resident block-seconds occupied by blocks whose holding sessions have all completed and that are never accessed again.

When re-reads are almost entirely within-session (SWE-bench, BrowseComp+, τ^2-Airline), completion preserves within one percentage point of the TTL's hits while holding 38–76% less dead state. In Qwen To-C and RAGPulse, it retains 15% and 13% more hits because the TTL prematurely expires paused sessions. Completion instead loses hits when shared blocks recur only after every session holding them has completed. RAGPulse's shared blocks recur quickly: only 6% of their recurrence gaps exceed ten sessions, so a session holding them is usually still live. In Qwen Code, 57% exceed ten sessions (Figure 9), and the TTL retains 37% more of all hits. Qwen To-B sessions are single requests, so every block is freed when its only request ends; all of its hits are cross-session, and the TTL retains 93% more of them.

This loss does not come from the lifecycle signal itself but from applying it to multi-session state, which has no completion boundary (§4.2). With a session ID, storage knows which blocks have been touched by more than one session (§4.4) and can exclude them from reclamation, applying lifecycle only to single-session state. The shared blocks it spares should then be retained according to their recurrence and capacity cost (§4.1) rather than their sessions' lifecycle.